



Submitted By:

Ali Zaman-CYS-35

Assignment:

LAB -4

Submitted To:

Sir. Mujtaba Kamal

For the Fulfillment Of:

FUNDAMENTALS OF PROGRAMMING

DEPARTMENT OF COMPUTER ENGINEERING

**UNIVERSITY OF ENGINEERING AND
TECHNOLOGY, LAHORE**

Task No.1:

Write a Python program that defines a user-defined function greet() which prints the message: “Welcome to Python Programming”. Call the function two times in the main program.

Solution:

```
def greet():  
    print("Welcome to Python Programming")  
  
# Main program  
greet()  
greet()
```

Output:

```
PS C:\Users\dell\AppData\Local\Programs\Microsoft VS Code> & C:  
Welcome to Python Programming  
Welcome to Python Programming  
PS C:\Users\dell\AppData\Local\Programs\Microsoft VS Code> |
```

Task No.2:

Write a Python program that defines a function displayName(name) which takes a name as an argument and prints: “Hello ”. Call the function by passing a name from the main program.

Solution:

```
def displayName(name):  
    print("Hello", name)  
  
# Main program  
displayName("Ali Zaman")
```

Output:

```
PS C:\Users\dell\AppData\Local\Programs\Microsoft  
Hello Ali Zaman  
PS C:\Users\dell\AppData\Local\Programs\Microsoft
```

Task No.3:

Write a Python program that defines a function addNumbers(a, b) which takes two numbers as parameters and returns their sum. Display the returned result in the main program.

Solution:

```
def addNumbers(a, b):  
    return a + b  
  
num1 = int(input("Enter first number: "))  
num2 = int(input("Enter second number: "))  
  
result = addNumbers(num1, num2)  
print("Sum =", result)
```

Output:

```
PS C:\Users\dell\AppData\Local\Programs\Microsoft VS Co  
Enter first number: 11  
Enter second number: 90  
Sum = 101  
PS C:\Users\dell\AppData\Local\Programs\Microsoft VS Co
```

Task No.4:

Write a Python program that defines a function square(num) which takes a number as input and returns the square of that number. Call the function and display the result.

Solution:

```
def square(num):  
    return num * num  
  
# Main program  
number = int(input("Enter a number: "))  
result = square(number)  
  
print("Square =", result)
```

Output:

```
PS C:\Users\dell\AppData\Local\Programs\Micro  
Enter a number: 7  
Square = 49  
PS C:\Users\dell\AppData\Local\Programs\Micro
```

Task No.5:

Write a Python program that defines a function `power(base, exponent=2)` using a default argument. If the exponent is not provided, the function should calculate the square of the base number. Call the function with and without providing the exponent.

Solution:

```
def power(base, exponent=2):  
    return base ** exponent  
  
# Main program  
  
# Calling without exponent (default = 2)  
print("Square:", power(5))  
  
# Calling with exponent  
print("Power:", power(5, 3))
```

Output:

```
PS C:\Users\dell\AppData\Local\Programs\Microsoft  
Square: 25  
Power: 125  
PS C:\Users\dell\AppData\Local\Programs\Microsoft
```

Task No.6:

Write a Python program that defines a function `student(name, age)` which prints the student's name and age. Call the function using keyword arguments.

Solution:

```
def student(name, age):  
    print("Student Name:", name)  
    print("Student Age:", age)  
  
# Main program (keyword arguments)  
student(name="Ali Zaman", age=19)
```

Output:

```
PS C:\Users\dell\AppData\Local\Programs\Microsoft  
Student Name: Ali Zaman  
Student Age: 19  
PS C:\Users\dell\AppData\Local\Programs\Microsoft
```

Task No.7:

Write a Python program that defines a function maximum(a, b, c) which takes three numbers as parameters and returns the largest number among them.

Solution:

```
def maximum(a, b, c):  
    if a >= b and a >= c:  
        return a  
    elif b >= a and b >= c:  
        return b  
    else:  
        return c  
  
print("Largest number:", maximum(15, 85, 99))
```

Output:

```
PS C:\Users\dell\AppData\Local\Programs\Microsoft VS  
Largest number: 99  
PS C:\Users\dell\AppData\Local\Programs\Microsoft VS
```

Task No.8:

Write a Python program that defines a function total(*numbers) using variable length arguments.

The function should calculate and print the sum of all numbers passed to it.

Solution:

```
def total(*numbers):  
    sum = 0  
    for num in numbers:  
        sum += num  
    print("Total =", sum)
```

Output:

```
PS C:\Users\dell\AppData\Local\Programs\VS Code\bin  
Total = 182  
PS C:\Users\dell\AppData\Local\Programs\VS Code\bin
```

Task No.9:

Write a Python program that defines a function average(*numbers) that accepts multiple numbers and calculates their average.

Solution:

```
def average(*numbers):
    sum = 0
    for num in numbers:
        sum += num

    avg = sum / len(numbers)
    print("Average =", avg)

# Main program
average(90, 24, 35, 40)
```

Output:

```
PS C:\Users\dell\AppData\Local\Program
Average = 47.25
PS C:\Users\dell\AppData\Local\Program
```

Task No.10:

Write a Python program that implements a menu-driven calculator using user-defined functions. Create the following functions:

add(a, b)

subtract(a, b)

multiply(a, b)

divide(a, b)

Display a menu to the user and call the appropriate function based on the user's choice.

Solution:

```
print("Calculator Menu")
print("1. Add")
print("2. Subtract")
print("3. Multiply")
print("4. Divide")

choice = int(input("Enter your choice (1-4): "))

num1 = float(input("Enter first number: "))
```

```
num2 = float(input("Enter second number: "))

if choice == 1:
    print("Result:", add(num1, num2))

elif choice == 2:
    print("Result:", subtract(num1, num2))

elif choice == 3:
    print("Result:", multiply(num1, num2))

elif choice == 4:
    print("Result:", divide(num1, num2))

else:
    print("Invalid choice")
```

Output:

```
Calculator Menu
1. Add
2. Subtract
3. Multiply
4. Divide
Enter your choice (1-4): 1
Enter first number: 5
Enter second number: 6
Result: 11.0
PS C:\Users\dell\AppData\Loc
```

```
Calculator Menu
1. Add
2. Subtract
3. Multiply
4. Divide
Enter your choice (1-4): 4
Enter first number: 7
Enter second number: 8
Result: 0.875
```